

AI Engineer (Agent Systems, Local RAG, Grounded AI Services)

AI engineer focused on grounded, tool-using systems. I build local RAG applications, AI services backed by live application data, and evaluation and approval boundaries for agent workflows. My backend foundation in Python, FastAPI, SQL, Docker, and API design supports reliable AI products.

PROFESSIONAL EXPERIENCE

FlyRank AI — Backend AI Engineering Intern | July 2026 – Present

- SafeBump (in progress): developing a capstone agent that scans dependency risk, works on an isolated branch, applies an upgrade, runs deterministic checks, and keeps or rolls back the change with a written reason.
- Designed human approval gates for major upgrades, push, and pull requests; implementation and evaluation results are not claimed before the scheduled build.

[SafeBump case study](#)

- Built and tested FastAPI assignments and designed a five-stage source-grounded research workflow with five real runs and documented human-review boundaries.

Microsoft Türkiye AI Innovators — Program Participant | June 2026 – July 2026

- Foundry Local RAG Assistant: built a fully local retrieval system where documents, embeddings, generation, and storage remain on the device; exposed one answer pipeline through CLI and web interfaces.
- Scored 8/10 on a development evaluation and 7/10 on a frozen blind set; preserved the methods, raw results, known limitations, and a 38-test suite.

github.com/bilgenurpala/local-rag-assistant

VBT Software — Backend Developer Intern | June 2026 – July 2026

- PetAdopt: co-built an adoption platform and owned the backend, database migrations and seed data, separate Claude AI service, Docker and CI setup, backend QA, documentation, and Flutter admin work.
- Grounded AI recommendations in read-only pet records instead of generated listings; the final Newman run covered 30 requests and 51 assertions with 0 failures.

github.com/bilgenurpala/pet-adopt

Software Persona — Backend Developer Intern | March 2026 – April 2026

- NovaStore: built FastAPI and MSSQL APIs for authentication and product workflows and integrated an AI assistant for customer and product interactions.

EDUCATION

- BSc, Management Information Systems (ongoing), Istanbul University, 2025–2028
- BSc, Finance, Eskisehir Osmangazi University, 2016–2023

SKILLS

- AI: RAG, tool calling, agent workflows, structured outputs, evaluation, prompt engineering, Claude API, Foundry Local
- Languages: Python, JavaScript
- Backend: FastAPI, Pydantic, SQLAlchemy, Node.js, Express.js, REST, OpenAPI
- Data: PostgreSQL, SQLite, MSSQL, MySQL
- Tools: Docker Compose, GitHub Actions, Git

CERTIFICATIONS

[Anthropic Academy: 20 completed courses and certificates](#)

- Claude with the Anthropic API
- MCP: Advanced Topics
- Introduction to Agent Skills
- Introduction to Subagents
- AWS Cloud Practitioner Essentials
- IBM & Kodluyoruz CyberStart

LANGUAGES

- Turkish — Native
- English — Intermediate